Telemetry-Aware Capacity Elasticity in a Three-Stage Service Chain

Mathematical model, independent verification and frozen experiments

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Thesis context: Optimal Resource Allocation in ORAN with SDN-Managed Orchestration.

Abstract

This computational study examines paid capacity elasticity under delayed telemetry and delayed actuation in a two-class, three-stage packet service chain. A finite-grid controller combines historical queue extrapolation, a dispersion margin and class reservations, with local deadline-based borrowing of unused capacity. Strong fixed comparators are selected using a separate calibration set. An additional locked evaluation comprises 480 replication/ablation runs and 120 activation-delay runs. The fresh replication reduces incremental abstract resource cost by 37.77–42.36% across the three stressed settings relative to the previously selected fixed expansions. However, prediction is not uniformly helpful, removing the margin reduces cost while increasing misses, and the nominal case exhibits unnecessary capacity activation. These findings support a cost–delay trade-off, not universal policy superiority. Independent packet replay, uncached optimizer enumeration and 37 unit tests verify the implemented abstraction. At a two-second actuation delay, the dynamic policy loses its cost advantage and has substantially more class-1 misses than fixed expansion. Physical O-RAN validation, state-of-the-art policy reproduction and a complete novelty review remain open.

Status and claim boundary

This is a reproducible technical study and manuscript foundation, not a submission-ready IEEE paper or validated O-RAN result. No NR physical layer, SDN protocol, RIC interface or calibrated CPU workload is implemented. All numeric parameters are synthetic choices. The study does not claim monetary or energy savings, URLLC guarantees, global optimality, a current Q1 classification or publication acceptance. No quantum component is included.

What this release adds

- A complete event-order contract, symbols and equations for the implemented model.
- Correctness propositions with explicit limits and an independent software check.
- Six controller variants, fresh seeds, command-delay stress tests and paired uncertainty.
- Negative results, source/parameter provenance and a concrete external-validation gate.

1 Context, closest work and contribution limits

Open Radio Access Network (O-RAN) exposes architectural interfaces and programmable control opportunities [1]. Timestamped telemetry creates an information-age problem rather than merely a sampling-rate problem [2]. Neither motivation demonstrates that a particular queue simulator is O-RAN compliant.

Here SDN means Software-Defined Networking, RIC means RAN Intelligent Controller, NR means New Radio, and URLLC means Ultra-Reliable Low-Latency Communication. FIFO means first-in, first-out and EDF means earliest-deadline-first. These terms identify motivations and scheduling concepts, not implemented protocol compliance.

MAREA develops radio-resource allocation around a Martingale-based delay analysis and multi-timescale control [3]. The current dispersion score is not a replacement for its mathematical delay bound. Kalntis et al. study online learning for virtualized base stations, including a meta-learning mechanism and trace-driven evaluation [4]. No such learning/regret result is established here. Joint radio, fronthaul and cloud orchestration is already studied in cell-free O-RAN [5]. Therefore "joint allocation" alone is not a new contribution.

OREO addresses xApp-based service orchestration and sharing [6], whereas the present queues represent packet service. Predictive shared AI/RAN orchestration is also prior art [7]. These problems are relevant context but not directly interchangeable benchmark implementations.

Prior work	Relevant distinction	Consequence for this study
MAREA	Radio delay analysis and two control timescales	Do not claim probabilistic delay guarantees from a queue proxy
Adaptive vBS allocation	Online learning and real-world-data evaluation	No claim to beat published learning methods without reproduction
Joint cell-free O-RAN	Coupled physical radio/fronthaul/cloud resources	Abstract stages do not demonstrate physical joint optimization
OREO	xApp sharing and service deployment	An xApp is not the same object as a packet in this model
Predictive AI/RAN	Forecasting and shared compute orchestration	Prediction itself is not a novelty claim

ColO-RAN provides an experimental framework for closed-loop xApps [8]. 5G-LENA supplies an end-to-end NR simulation route [9], with documented calibration against reference scenarios [10]. Neither was executed for these results. Their existence identifies a route to external validation, not a validation certificate for the current code.

Potential research question: How should information refresh and paid resource elasticity account for pending commands when both sensing and actuation are delayed? The present experiment explores this interaction but does not yet establish a substantial novel theoretical or empirical contribution. The three planned papers cannot be obtained by renaming the same simulation outputs.

2 System model, units and event order

Let $i \in \{1, 2, 3\}$ index stages and $s \in \{0, 1\}$ index classes. Time t is discrete. $A_s(t)$ counts new packets, $Q_{is}(t)$ is the start-of-slot queue before external arrivals and $S_{is}(t)$ is the number served in that slot. Base capacity $c_i(t)$ and active expansion $e_i(t)$ are in packets/slot. The evolution is

$$Q_{1s}(t+1) = Q_{1s}(t) + A_s(t) - S_{1s}(t), \tag{1}$$

$$Q_{is}(t+1) = Q_{is}(t) - S_{is}(t) + S_{i-1,s}(t), \quad i = 2, 3. \tag{2}$$

Stage 1 may serve newly arrived packets. Stages 2 and 3 may serve only packets present before upstream service in that slot. Each packet consumes one service unit per stage. Class deadlines are $D_0 = 15$ and $D_1 = 40$ slots.

Parameter	Fixed design choice
Slot mapping	20 ms, synthetic rather than an NR slot
Traffic	Poisson rates $(0.9, 2.1)$ packets/slot, burst rates $(1.8, 3.8)$
Burst	$180 \le t \bmod 600 < 290$, stressed settings only
Base service	$5 + \xi$ with $\xi \in \{-1, 0, 0, 1\}$ sampled uniformly by list position
Stressed stage	Capacity 3 during $170 \le t \bmod 600 < 310$
Extra service	Radio-like stage 0, transport/compute-like stages each 0–3
Report request interval/history	5 slots / at most 20 previous slots
Report delay/loss	3 slots and 0.05, or 9 slots and 0.15
Activation delay	10 slots by default, 1–100 in sensitivity study
Initial priors	Queue 0, arrivals $(0.9, 2.1)$, service $(2.5, 2.5)$, variance $(1, 1)$
Measured cohort	Birth slots 300–3299, simulation through slot 3339
Switching coefficient	$\eta = 0.25$, abstract cost units

Slot order. Apply due commands, deliver due reports, insert external arrivals, request a report and compute a new command when $t \bmod 5 = 0$, serve stages in order 3, 2, 1, update local history and check packet conservation. The report snapshot is after external arrivals and before service. Loss is determined by request index. Timestamps refer to measurement time.

Local scheduler. Largest-remainder quotas split $c_i(t) + e_i(t)$ using $(\alpha, 1 - \alpha)$. Unused quotas are borrowed by the next queued packet with earliest absolute deadline. Within-class service is FIFO. This is quota-first scheduling, not a claim of globally optimal EDF scheduling.

3 Controller formulation and execution contract

For the last received report timestamp τ_i, queue q_{is}, historical arrival mean λ_{is}, service mean μ_{is} and net-flow variance v_{is}, the activation-time extrapolation and demand proxy are

$$\widehat{q}_{is}(t+L) = \max\{0, q_{is} + (t+L-\tau_i)(\lambda_{is} - \mu_{is})\}, \tag{3}$$

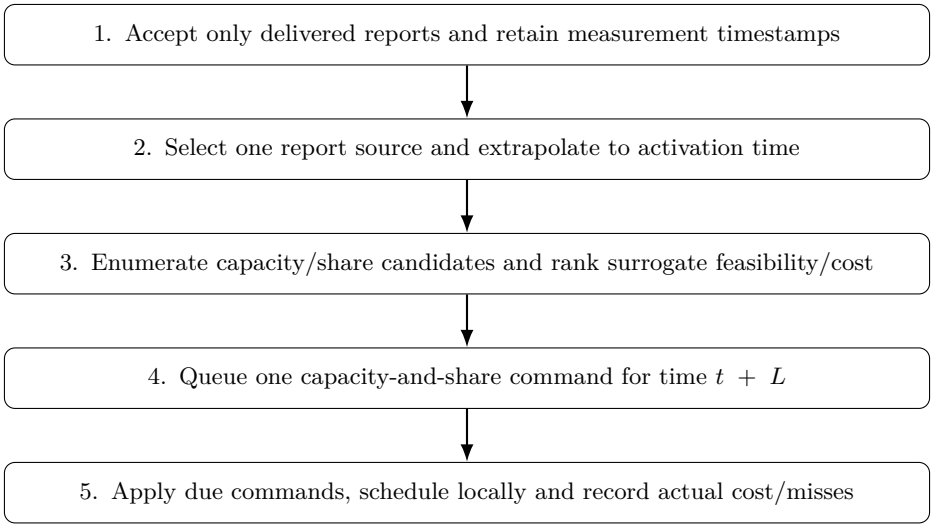
$$z_{is} = \widehat{q}_{is}(t+L) + 5\lambda_{is} + \sqrt{(t+L-\tau_i)v_{is}} + 1. \tag{4}$$

The candidate grid is $\mathcal{X} = \{0\} \times \{0,1,2,3\}^2$ and $\mathcal{A} = \{k/20 : k = 1, \ldots, 19\}$. Let $r_{is}(x,\alpha)$ denote integer quotas from reported base capacity plus candidate expansion. Define

$$\rho_s(x,\alpha) = \frac{1}{D_s} \sum_{i=1}^{3} \frac{z_{is}}{\max\{0.001, r_{is}(x,\alpha)\}}, \tag{5}$$

$$g(x) = \sum_i x_i + \eta \sum_i |x_i - x_i^{\text{prev request}}|. \tag{6}$$

A candidate with $\rho_{\max} = \max_s \rho_s \le 1$ is *surrogate feasible*. Lexicographic scores are $(0, g, \rho_{\max}, \sum_s \rho_s, k)$ if feasible, otherwise $(1, \rho_{\max}, g, \sum_s \rho_s, k)$. All 304 candidates are enumerated. Exact ties retain the first expansion pair in ascending order.



The full telemetry priority is

$$P_i = \sum_s \frac{\sqrt{(t-\tau_i)v_{is} + 1}}{D_s} \left(1 + \frac{\widehat{q}_{is}(t)}{\max\{1, c_i^{\text{report}}\}}\right) + 0.01(t - \tau_i).$$

The maximum is refreshed. The age term is not a starvation proof. The square-root margin is not statistically calibrated. The 0.001 quota floor is a numerical penalty, not real service. Commands are not cancelled or coalesced, and the predictor does not propagate the pending-command pipeline.

4 Correctness statements and their limits

Proposition 1: packet conservation. Cumulative generated packets equal cumulative completed packets plus queued packets at every slot. Summing (1)–(2) over stages cancels internal transfers. The simulator checks this identity per slot.

Proposition 2: work conservation. Total service at a stage is the minimum of active capacity and eligible queued packets. Initial quotas cannot overserve a class. Borrowing uses every remaining unit until either the capacity or all eligible queues are exhausted. This does not prove deadline feasibility.

Proposition 3: minimum packet latency. For birth slot b_p and completion slot f_p, $f_p - b_p + 1 \geq 3$. The packet must cross three stages and cannot cross two in one slot. The independent replay also checks this boundary.

Proposition 4: finite-grid optimality. The chosen candidate minimizes the stated lexicographic score over $\mathcal{X} \times \mathcal{A}$. After each iteration the incumbent is the minimum of the candidates examined so far. Induction over the finite enumeration proves the statement. It is not continuous optimality, a stochastic-control theorem or optimality of the total observed cost.

Proposition 5: a necessary capacity cut. All packets born in slots $[a, b)$ that traverse stage i and have deadlines at most $D_{\max}$ must cross that stage before their end-to-end deadline. Even granting the enlarged window $[a, b + D_{\max})$, a necessary condition for zero misses is

$$\sum_{t=a}^{b-1} \sum_{s} A_s(t) \leq \sum_{t=a}^{b+D_{\max}-1} \big(c_i(t) + e_i(t)\big).$$

Exceeding the right-hand side proves some deadline failures are unavoidable for those arrivals and capacities. Satisfying the condition does not prove schedulability. A failed zero-miss condition does not rule out a positive miss target.

Complexity. Direct enumeration is $O(MKIS)$ for M expansion combinations, K shares, I stages and S classes. Here $(M, K, I, S) = (16, 19, 3, 2)$. Caching removes repeated stage calculations, but expansion-grid growth remains exponential if more independently expandable stages are added. No scalability claim beyond this small model is made.

Actual cost. Across the entire simulation, including warmup and followup,

$$J = \sum_{t=0}^{T-1} \sum_{i} e_i(t) + \eta \sum_{t \in \mathcal{T}_{\text{apply}}} \sum_{i} |e_i(t) - e_i(t-1)|.$$

Static capacity is preprovisioned, with no initial switching charge but full capacity-time cost. The controller's one-decision score g is not J. Base infrastructure costs, power and financial prices are absent.

5 Evaluation design and statistical interpretation

The earlier calibration uses seeds 4000–4004 and all 16 fixed capacity pairs. The cheapest fixed pair whose two calibration mean miss rates are at most 1% is selected. The resulting choices are $(0, 0)$ for nominal, $(2, 0)$ for transport, $(0, 2)$ for compute and $(2, 0)$ for delayed/lost transport. The 1% threshold is only a comparator-selection rule, not a proven service-level agreement.

The earlier held-out evaluation uses seeds 5000–5019. This release uses a new local protocol lock, written before seeds 6000–6019 were evaluated. Its SHA-256 is stored with outputs. It is not externally preregistered. Six variants over four settings and 20 seeds give 480 runs. Six command delays over ten different seeds 7000–7009 and two policies give 120 additional runs.

Variant	Single controlled difference from full dynamic
Round-robin	Requests rotate through three stages
Oldest-first	Requests choose the oldest delivered report
No prediction	Uses reported queue directly, still retains the dispersion margin
No margin	Removes the square-root term, retains queue extrapolation
Selected fixed	Locks capacity at prior calibrated values, retains delayed share control

The measured class rate is the number of late-completed plus unfinished measured packets divided by all arrivals of that class in the measured cohort. The 40-slot tail ensures every measured deadline has elapsed. Background arrivals continue during that tail. Completed-only percentiles are not primary endpoints because they can hide unfinished packets.

Seed, not packet, is the independent replication unit. Percentile bootstrap intervals use 10,000 seed-level resamples, pairing policies by seed [11]. The intervals are exploratory, especially near zero events. For four settings and three endpoints, exact paired sign tests omit ties and test whether positive and negative paired differences are equally likely. All 12 p-values receive Holm correction as one family [12]. These tests are not tests of mean equivalence, noninferiority or an SLA.

If all 20 independent runs have no misses, the exact one-sided 95% upper bound on the probability that a new same-distribution run has any miss is $1 - 0.05^{1/20} = 13.91\%$. This is a run-level statement, not a packet failure probability. The bound follows from $(1 - p)^{20} = 0.05$.

6 Fresh-seed results: all settings and variants

All values below are seed means for seeds 6000–6019. Miss rates are percentages. Cost is incremental and abstract. All six variants are retained, including negative ablation outcomes.

Setting	Variant	Class 0 (%)	Class 1 (%)	Cost
Nominal	Full dynamic	0.0000	0.0000	142.31
Nominal	Round-robin	0.0000	0.0000	317.43
Nominal	Oldest-first	0.0000	0.0000	281.38
Nominal	No prediction	0.0000	0.0000	109.00
Nominal	No margin	0.0000	0.0000	0.00
Nominal	Selected fixed	0.0000	0.0000	0.00
Transport stress	Full dynamic	0.0000	0.0998	3863.60
Transport stress	Round-robin	0.0000	0.1168	4362.56
Transport stress	Oldest-first	0.0000	0.0876	4290.81
Transport stress	No prediction	0.0000	0.1093	3855.54
Transport stress	No margin	0.0325	0.4295	2654.43
Transport stress	Selected fixed	0.0000	0.1249	6680.00
Compute stress	Full dynamic	0.0000	0.1018	3850.61
Compute stress	Round-robin	0.0000	0.1202	4341.12
Compute stress	Oldest-first	0.0000	0.0896	4334.00
Compute stress	No prediction	0.0000	0.1107	3880.66
Compute stress	No margin	0.3887	0.4248	2602.40
Compute stress	Selected fixed	0.0000	0.1344	6680.00
Delayed/lost reports	Full dynamic	0.0000	0.1394	4156.99
Delayed/lost reports	Round-robin	0.0125	0.1005	4706.84
Delayed/lost reports	Oldest-first	0.0125	0.1148	4806.21
Delayed/lost reports	No prediction	0.0000	0.1046	4324.11
Delayed/lost reports	No margin	1.0017	1.6813	2574.16
Delayed/lost reports	Selected fixed	0.0000	0.1372	6680.00

For stressed settings, fixed cost is 6680 units. Fresh-seed dynamic savings are Transport stress: 42.16%, Compute stress: 42.36%, Delayed/lost reports: 37.77%. These estimates supplement, rather than replace or pool with, the earlier 38.82–43.28% savings from seeds 5000–5019. Comparison against a permanent maximum expansion is weaker and must not headline the study.

7 Component evidence and negative results

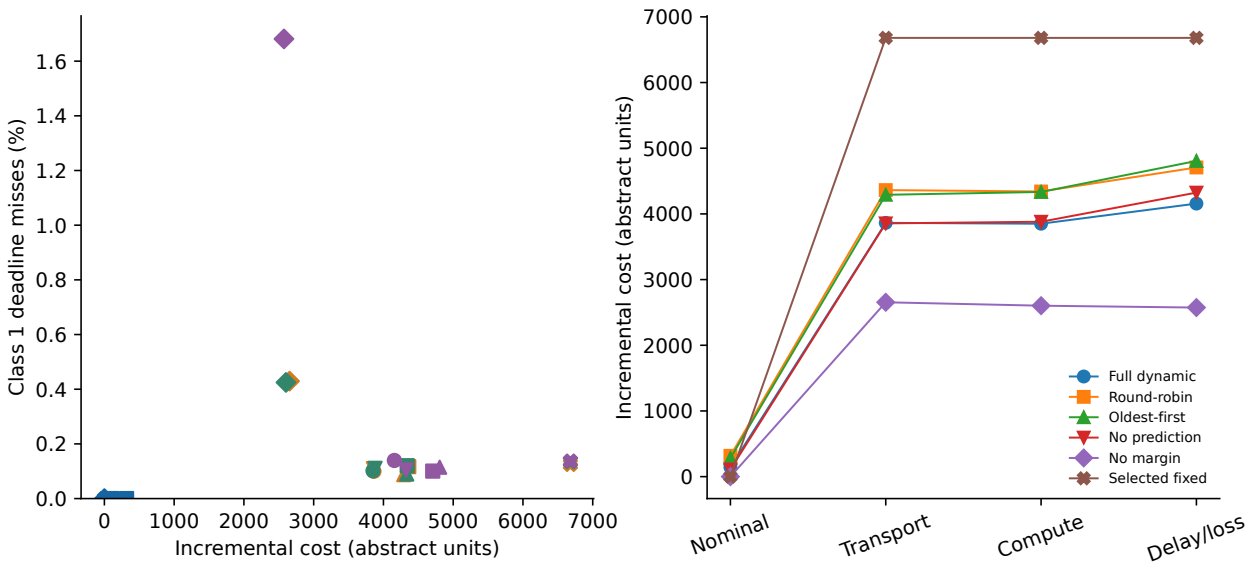


Figure 1. Cost and deadline outcomes across all variants. Scatter colors identify nominal (blue), transport (orange), compute (green) and delayed/lost reports (purple). Marker shapes identify methods, as in the right-panel legend. Both panels are descriptive seed means. The numeric table on the preceding page includes both classes and avoids hiding small class-0 rates.

Margin trade-off. Removing the margin substantially reduces resource cost but raises misses under stress. It removes needless nominal expansion in this sample. This supports a conservatism trade-off, not a calibrated reliability guarantee.

Prediction is not uniformly beneficial. Under delayed/lost reports, removing extrapolation lowers the observed class-1 miss rate at higher cost. Under ordinary transport stress it slightly lowers cost while increasing misses. The present evidence does not justify saying that prediction improves every case.

Telemetry priority is not universally dominant. Round-robin and oldest-first cost more in the stressed settings, but can have lower class-1 miss rates. In the delayed/lost setting they also show small class-0 misses absent in the full variant. The two-class trade-off must remain visible.

Nominal overhead. Fixed zero expansion attains the same observed zero misses as full dynamic with no extra resource cost. Relative savings against a zero-cost comparator are undefined, not 0% or 100%. This failure case should inform any future redesign, which must be tested on a new evaluation set.

8 Activation-delay sensitivity

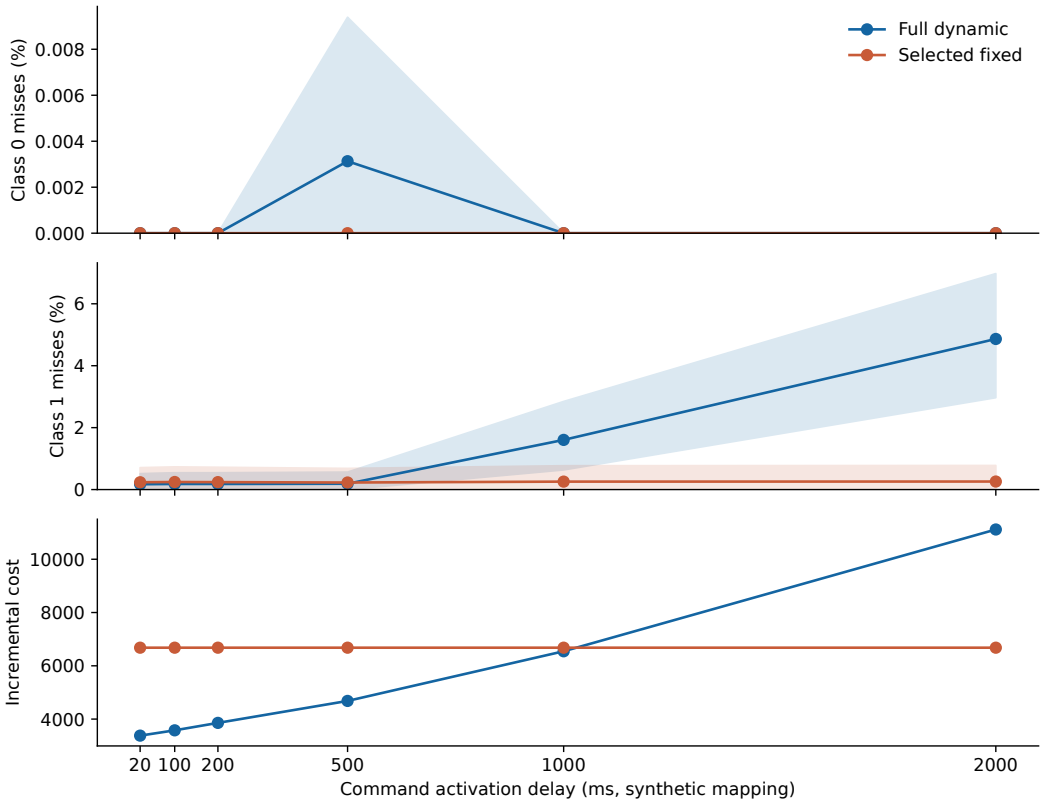


Figure 2. Transport stress with report delay 3 slots and loss 0.05. Shading shows exploratory 95% seed-bootstrap intervals for miss rates. Ten seeds per point are separate from the primary replication. Fixed expansion is preprovisioned, but its share updates retain the same command delay.

Delay (ms)	Variant	Class 0 (%)	Class 1 (%)	Cost
20	Full dynamic	0.0000	0.1715	3379.07
20	Selected fixed	0.0000	0.2359	6680.00
100	Full dynamic	0.0000	0.1811	3577.88
100	Selected fixed	0.0000	0.2442	6680.00
200	Full dynamic	0.0000	0.1811	3857.35
200	Selected fixed	0.0000	0.2401	6680.00
500	Full dynamic	0.0031	0.1879	4681.50
500	Selected fixed	0.0000	0.2277	6680.00
1000	Full dynamic	0.0000	1.6042	6545.60
1000	Selected fixed	0.0000	0.2565	6680.00
2000	Full dynamic	0.0000	4.8641	11116.77
2000	Selected fixed	0.0000	0.2593	6680.00

The delay sweep is an exploratory stress test, not measured RIC/SDN or cloud activation latency. Pending commands are applied in order rather than superseded. At 2 s, dynamic class-1 misses are 4.8641% versus 0.2593% for fixed expansion, while costs are 11116.78 versus 6680.00. This is a clear failure region of the present design, not an outlier to remove. It motivates pending-command-aware control and a new independent evaluation after any redesign.

9 Paired uncertainty, tests and verification

Differences are full dynamic minus selected fixed. Negative cost means lower incremental cost. Class differences are percentage points (pp). Confidence intervals concern seed-mean differences. Holm p-values concern paired direction, not the same estimand as a mean-difference test.

Setting	Endpoint	Difference	95% bootstrap interval	Holm p
Nominal	Class 0 (pp)	0.0000	[0.0000, 0.0000]	1.000
Nominal	Class 1 (pp)	0.0000	[0.0000, 0.0000]	1.000
Nominal	Cost	142.3125	[119.8747, 165.6144]	0.000023
Transport stress	Class 0 (pp)	0.0000	[0.0000, 0.0000]	1.000
Transport stress	Class 1 (pp)	-0.0251	[-0.0754, 0.0000]	1.000
Transport stress	Cost	-2816.4000	[-2890.7250, -2741.5741]	0.000023
Compute stress	Class 0 (pp)	0.0000	[0.0000, 0.0000]	1.000
Compute stress	Class 1 (pp)	-0.0326	[-0.0978, 0.0000]	1.000
Compute stress	Cost	-2829.3875	[-2906.1250, -2744.0103]	0.000023
Delayed/lost reports	Class 0 (pp)	0.0000	[0.0000, 0.0000]	1.000
Delayed/lost reports	Class 1 (pp)	0.0022	[-0.1243, 0.1310]	1.000
Delayed/lost reports	Cost	-2523.0125	[-2613.8394, -2425.0434]	0.000023

Intervals that include zero do not establish equivalence. All-zero bootstrap intervals do not prove zero population risk. No noninferiority margin was specified, so no noninferiority claim is made.

Independent software verification

The packet reference uses simultaneous outgoing lists and birth-time FIFO queues, whereas the production model uses downstream-first deque updates. Across 18 scenario/policy checks, arrivals, late completions, unfinished packets, miss rates, backlog and completed-delay percentiles agree exactly. This reference replays applied actions, so it validates the data plane conditional on those actions, not the whole controller implementation.

An independently written, uncached optimizer checks 30 random report states against the cached implementation. Chosen expansions/shares agree, and numerical ratios agree to 12 decimal places. Separate tests cover deadline boundaries, zero load, zero capacity, future-trace noninterference and invalid delays. The complete suite has 37 passing unit-test methods. Six regression comparisons confirm unchanged default behavior against the prior engine.

Whole-output validation checks all 600 new records, the locked protocol hash, seed separation, report budgets, cost identities, capacity limits, measured-cohort rates and the 12-test Holm adjustment. These are software checks, not physical model validation. Raw records and scripts are included in the reproducibility archive.

10 Publication gate and next experiment

IEEE TNSM explicitly welcomes theoretical and applied network/service-management contributions [13]. Thus hardware is not universally mandatory. However, neither scope fit nor clean formatting establishes originality or acceptance. Current journal quartiles were not verified in this study.

Requirement	Current state	Evidence still required
Mathematical model	Explicit and matched to code	Substantial new result beyond finite-grid correctness
Software verification	37 tests and independent checks	Broader scale/topology validation after model extension
Physical calibration	Not available	Radio/link/CPU model with documented units and provenance
External validation	Not executed	Calibrated network simulator or independent measured traces
Published baselines	Context reviewed, not reproduced	Faithful compatible implementations and matched assumptions
Literature depth	13 focused sources	Complete closest-work review, including the requested 40 relevant references
Three-paper program	One abstract model only	Distinct controllers/questions and separate evidence for each paper

Required source inputs. The approved proposal and the student's current simulator/configuration or measured traces are needed to align a physical study with the thesis. If none exists, a deployment scenario must be selected for a calibrated system-level simulator. Necessary fields include bandwidth, channel, topology, users, payload sizes, transport links, compute workload, deadlines, telemetry overhead and resource activation mechanisms.

Proposed next validation, not completed work. Pin a documented external calibration case and reproduce its reference metrics first. Define acceptable differences before viewing outputs. Introduce the controller without changing the calibrated data-plane defaults. Reproduce the closest compatible published baseline, freeze fresh test traces, and report all differences and failure cases.

Three-paper separation. A telemetry/elasticity paper needs stronger novelty and validation. An asynchronous-control paper needs a pending-action-aware controller and a new safety/feasibility result. A recovery paper needs explicit failures, recovery actions and recovery-time constraints. The present burst tests do not constitute a fault-recovery experiment.

Conclusion. The artifact establishes a reproducible synthetic trade-off between paid capacity, delayed information and deadline misses. It also identifies negative outcomes and uncontrolled assumptions. These results are sufficient for technical review and model development, not for claiming completed Q1-ready O-RAN papers.

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